Notes on PAC-Bayes and Online Bayesian Inference

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1 Preliminaries

This note is my self-study notes based on several references, including Alquier [2021, 2024], Chérief-Abdellatif et al. [2019].
In this section, we introduce several useful technical lemmas. (the good old trick)

Lemma 1 (Donsker-Varadhan's Variational Formula). *For any measurable bounded function $h : \Theta \to \mathbb{R}$,*

$$\log \mathbb{E}_{\theta\sim\pi}[e^{h(\theta)}] = \sup_{p\in\mathcal{S}(\Theta)} \left[\mathbb{E}_{\theta\sim p}[h(\theta)] - KL(p\|\pi)\right]$$

Lemma 2 (Hoeffding's inequality). *Let $X_1, \ldots, X_n$ be independent random variables such that*

$$a_i \le X_i \le b_i \qquad \text{almost surely for each } i = 1, \ldots, n.$$

Then, for any $t > 0$,

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mathbb{E}[X_i]) \ge t\right) \le \exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

In particular,

$$\mathbb{P}\left(\left|\sum_{i=1}^{n}(X_i - \mathbb{E}[X_i])\right| \ge t\right) \le 2\exp\left(-\frac{2t^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

2 Online Bayesian Inference

For Bayesian learning, thanks to Donsker-Varadhan's variational formula[e.g. See Corollary 2.3 of Alquier [2024]], we can view bayesian update as following:

$$\begin{aligned}
&p_{t+1}(\theta) \propto \exp(-\eta \sum_{i=1}^{t} \ell_t(\theta))\pi(\theta) \\
&\iff p_{t+1}(\theta) \propto \exp(-\eta\, \ell_t(\theta))p_t(\theta) \\
&\iff p_{t+1}(\theta) = \operatorname*{arg\,min}_{p\in\mathcal{S}(\Theta)} \left\{ \mathbb{E}_{\theta\sim p}\left[\sum_{i=1}^{t} \ell_i(\theta)\right] + \frac{KL(p\|\pi)}{\eta} \right\}.
\end{aligned}$$

For some reasons, the family $\mathcal{S}(\Theta)$ may be too general or computationally intractable. Thus, we restrict it to a tractable subfamily parameterized by ϕ, known as the variational family:

$$\mathcal{F} := \{q_\phi \in \mathcal{S}(\Theta) : \phi \in \Phi\}.$$

A similar argument applies here as in the case of $\mathcal{S}(\Theta)$. We simply replace $\mathcal{S}(\Theta)$ with $\mathcal{F}$ and perform a first-order expansion. Define

$$\bar{L}_t(\phi) := \mathbb{E}_{\theta\sim q_\phi}\left[\ell_t(\theta)\right].$$

Using a first-order approximation around ϕ_t, we have

$$\bar{L}_t(\phi) \approx \bar{L}_t(\phi_t) + (\phi - \phi_t)^\top \nabla_\phi \bar{L}_t(\phi_t).$$

Ignoring constants independent of ϕ, the update becomes

$$\phi_{t+1} = \arg\min_{\phi\in\Phi}\left\{\phi^\top \nabla_\phi \bar{L}_t(\phi_t) + \frac{KL(q_\phi \| q_{\phi_t})}{\eta}\right\}.$$

which is known as Streaming Variational Bayes [Broderick et al., 2013].

2.1 Regret Bounds for Online Bayesian Inference

The main idea of the proof is quite similar to that of the Hedge algorithm (in my opinion..), as it is based on exponential reweighting. And the convexity assumption in ℓ_t and the use of posterior mean $\hat{\theta}_t = \mathbb{E}_{\theta\sim p_t}(\theta)$ play a crucial role in the proof.

$$\begin{aligned}\exp[-\eta\ell_t(\hat{\theta}_t)] &= \exp[-\eta\ell_t(\mathbb{E}_{\theta\sim p_t}(\theta)] \\ &\geq \exp[-\eta\mathbb{E}_{\theta\sim p_t}(\ell_t(\theta))]\end{aligned}$$

where the first inequality follows from Jensen's inequality.
Note that by Hoeffding's inequality,

$$\begin{aligned}&\mathbb{E}_{\theta\sim p_t}[\exp(-\eta\ell_t(\theta)) - \mathbb{E}_{\theta\sim p_t}(\ell_t(\theta))] \leq \exp(\eta^2 B^2/8) \\ &\Rightarrow \mathbb{E}_{\theta\sim p_t}[\exp(-\eta\ell_t(\theta))] \leq \exp(-\eta\mathbb{E}_{\theta\sim p_t}(\ell_t(\theta)) + \eta^2 B^2/8) \\ &\Rightarrow \exp(-\eta\mathbb{E}_{\theta\sim p_t}(\ell_t(\theta))) \geq \mathbb{E}_{\theta\sim p_t}[\exp(-\eta\ell_t(\theta) - \eta^2 B^2/8)]\end{aligned}$$

Thus,

$$\exp[-\eta\ell_t(\hat{\theta}_t)] \geq \mathbb{E}_{\theta\sim p_t}[\exp(-\eta\ell_t(\theta) - \eta^2 B^2/8)]$$

So, by rearranging this term provide us

$$\ell_t(\hat{\theta}_t) \leq \frac{\eta^2 B^2}{8} - \frac{1}{\eta}\log \mathbb{E}_{\theta\sim p_t}\exp(-\eta\ell_t(\theta)) \tag{1}$$

Recall that the generalized bayes (or Gibbs posterior update) can be expressed as

$$p_t = \frac{\exp(-\eta\sum_{i=1}^{t-1}\ell_i(\theta))\pi(\theta)}{N_t}$$

where N_t is normalization constant given by

$$N_t = \mathbb{E}_{\theta\sim\pi}[\exp(-\eta\sum_{i=1}^{t-1}\ell_i(\theta))]$$

Then, one can observe that

$$\log \mathbb{E}_{\theta\sim p_t}\exp(-\eta\ell_t(\theta)) = \log(N_{t+1}/N_t) \tag{2}$$

Then, summing 1 over t and using 2, we obtain

$$\begin{aligned}\sum_{t=1}^{T} \ell_t(\hat{\theta}_t) &\leq \frac{\eta B^2 T}{8} - \frac{1}{\eta}\sum_{t=1}^{T} \log(N_{t+1}/N_t) \\ &= \frac{\eta B^2 T}{8} - \frac{1}{\eta}\log(N_{t+1}/N_t) \\ &= \frac{\eta B^2 T}{8} - \frac{1}{\eta}\log(\mathbb{E}_{\theta\sim\pi}[\exp(-\eta \sum_{t=1}^{T}\ell_t(\theta)]\end{aligned}$$

Then we can easily recognize the form of the Donsker–Varadhan formula!

$$\sum_{t=1}^{T} \ell_t(\hat{\theta}_t) \leq \frac{\eta B^2 T}{8} + \inf_{p\in\mathcal{S}(\Theta)} \left\{ \mathbb{E}_{\theta\sim p}\left[\sum_{t=1}^{T}\ell_t(\theta)\right] + \frac{KL(p\|\pi)}{\eta} \right\}$$

2.2 Regret Bounds for Online Variational Inference

Since several papers, such as Chérief-Abdellatif et al. [2019], Alquier [2024], already provide very clear proofs, I will only describe my intuition and emphasize the key ideas behind the argument.
The proof heavily relies on two ingredients!

(i) the "convex" loss $\ell_t(\theta)$,

(ii) the posterior "mean" estimator $\hat{\theta}_t = \mathbb{E}_{\theta\sim p_t}(\theta)$.

For example, in online variational inference, by the convexity of $\ell_t(\theta)$, we have

$$\sum_{t=1}^{T} \ell_t(\hat{\theta}_t) - \inf_{\phi\in\Phi} \mathbb{E}_{\theta\sim q_\phi}\left[\sum_{t=1}^{T}\ell_t(\theta)\right] \leq \mathbb{E}_{\theta\sim p_t}\left[\sum_{t=1}^{T}\ell_t(\theta)\right] - \inf_{\phi\in\Phi} \mathbb{E}_{\theta\sim q_\phi}\left[\sum_{t=1}^{T}\ell_t(\theta)\right].$$

This shows that it suffices to prove that no regret with respect to the mean loss implies no regret with respect to the cumulative loss, giving a direct connection between the two perspectives.

3 Still studying — notes will be added later

My plans for further reading are:

1. Connections between Online Conformal Prediction and PAC-Bayes. I find the idea quite interesting.

References

Pierre Alquier. Non-exponentially weighted aggregation: regret bounds for unbounded loss functions. In *International Conference on Machine Learning*, pages 207–218. PMLR, 2021.

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